FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2015

CONCEPT RECAPITULATION TEST - I

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 10)** contains 10 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

(ii) **Section-C (01 – 10)** contains 10 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

(One or More than One Options Correct Type)

This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

1. Initially two particles *A* and *B* are present at (0, 0) and (*d*, 0) respectively. They start moving with speed $v_A = v\hat{i} + v\hat{j}$ and $v_B = -v\hat{j}$. If *R* is magnitude of relative separation between them and T_0 be the time when separation between them is minimum, then

   (A) $T_0 = \dfrac{d}{5v}$ (B) $R_{\min} = \dfrac{2d}{\sqrt{5}}$

   (C) Graph of *R* versus time is straight line (D) Graph of *R* versus time is circle.

2. Four forces act on a point object. The object will be in equilibrium if
   (A) all of them are in the same plane
   (B) they are opposite to each other in pair
   (C) the sum of *x*, *y* and *z* components of force is zero separately
   (D) they form a closed figure of 4 sides

3. A person of mass 70 kg jump from a 3.0 m height. Then
   (A) impulse on the man by ground will be 539 N-s
   (B) average force on man if he land with stiffed leg and body moves by 1.0 cm during impact is 2.1×10^5 N
   (C) average force on man feet by ground if he land with bent legs and body moves by 50 cm is 4.2×10^3 N
   (D) average force on man feet by ground if he land with bent legs and body moves by 50 cm is (4.9×10^3 N)

4. A mass *m* of radius *r* is rolling horizontally without any slip with a linear speed *v*. It then rolls up to a height given by $\dfrac{3}{4}\dfrac{v^2}{g}$

   (A) the body is identified to be a disc or a solid cylinder
   (B) the body is a solid sphere
   (C) MOI of the body about instantaneous axis of rotation is $\dfrac{3}{2}mr^2$
   (D) MOI of the body about instantaneous axis of rotation is $\dfrac{2}{5}mr^2$

Space for Rough work

5. A particle initially at rest is displaced by applying a non conservative force F in a uniform gravitational field. In the process following physical quantities associated with the particle are measured

$\Delta U =$ change in gravitational potential energy

$\Delta K =$ change in kinetic energy

$\Delta W_1 =$ work done by the force F

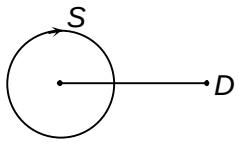
$\Delta W_2 =$ work done by the gravitational force

(A) $\Delta W_2 = -\Delta U$ (B) $\Delta K = \Delta W_1 + \Delta W_2$

(C) $\Delta K + \Delta U = \Delta W_1 + \Delta W_2$ (D) $\Delta W_1 = \Delta W_2$

6. An object is floating in a liquid, kept in a container. The container is placed in a lift. Choose the correct option(s)
(A) Buoyant force increases as lift accelerates up
(B) Buoyant force decreases as lift accelerates up
(C) Buoyant force remains constant as lift accelerates
(D) The fraction of solid submerged into liquid does not change

7. A source of sound moves along a circle of radius 2 m with constant angular velocity 40 rad/s. Frequency of the source is 300 Hz. A detector is kept at some distance from the circle in the same plane of the circle (as shown in figure). Which of the following is not the possible value of frequency registered by the detector? (Speed of sound = 320 m/s)



(A) 250 Hz (B) 360 Hz
(C) 410 Hz (D) 220 Hz

8. Two gases have same initial pressure, volume and temperature. They expand to same final volume, one adiabatically and the other isothermally.
(A) the final temperature is greater for isothermal process
(B) the final temperature is lesser for isothermal process
(C) the work done by the gas is greater for isothermal process
(D) the work done by the gas is greater for adiabatic process

Space for Rough work

9. A charged particle q is placed at a distance d from the centre of conducting sphere of radius $R(<d)$, then in static condition at the centre of sphere

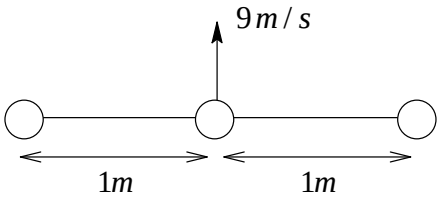
(A) magnitude of electric field due to induced charge is $\frac{kq}{d^2}$

(B) magnitude of electric field due to induced charge is $\frac{kq}{R^2}$

(C) magnitude of electric field due to induced charge is zero

(D) magnitude of electric field due to charge q is $\frac{kq}{d^2}$ (where $k = 1/4\pi\varepsilon_0$)

10. Three identical small balls of same mass 1 kg are placed on a horizontal smooth table. The velocity give to middle ball parallel to the surface is 9 m/s. Then:



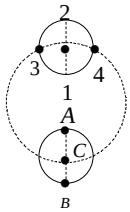
(A) the speed of colliding balls just before collision is 5 m/s
(B) the speed of any of the balls when they collide is 9 m/s
(C) the tension in the string connected between balls just before the two balls collide is 5 newton
(D) the tension in the string connected between balls just before the two balls collide is 9 newton

Space for Rough work

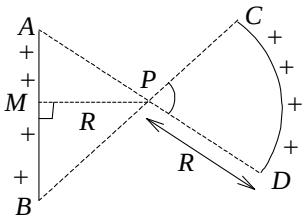
SECTION – C
(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

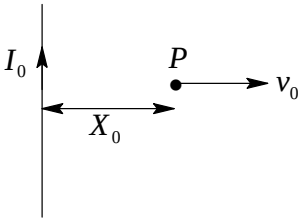
1. A ball translates on a horizontal circular path as shown in the figure. When the ball reaches the diametrically opposite point. The position of point A in shown by point 1, 2, 3 and 4. Find the correct position of point A:



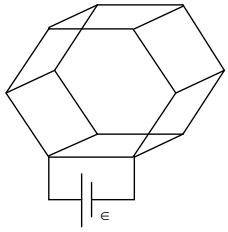
2. A rod of finite length having linear charge density λ subtends an angle of $15°$ at a point P which is also the centre of curvature of a circular arc having same linear charge density λ as shown in figure. If the net electric field at the origin is $\frac{bK\lambda}{R}$. Find the value of b:



3. A charge particle having specific charge α is initially located at point P at a distance of X_0 from an infinite current carrying wire as shown in figure. The maximum distance from the wire the charge will go, if it is given velocity v_0 perpendicular to the wire is $X_0 e^{\frac{b\pi v_0}{\alpha\mu_0 I_0}}$. Find the value of b:

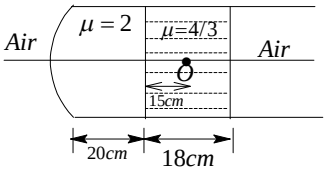


4. An electrical network is formed by two hexagon and joining grid points with wires. The resistance of each side and the connecting wires is R. An ideal battery of emf E is joined as shown in figure. Find the current passing through the battery. Consider the wire connected with the cell is ideal. [take $E = 11$ Volt ; $R = 6\,ohm$]

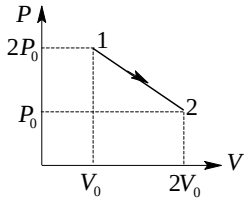


Space for Rough work

5. A point object is placed at the midpoint in water having boundary width of 30 cm and other medium having refractive index 2 is having boundary width 20 cm as shown in figure. The position of image of object O as seen from air through the curved part is 10k. Find the value of k…….

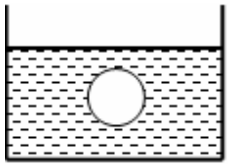


6. An ideal gas undergoes a process shown in the P-V diagram. If the change in heat content of the gas in this process is $\frac{K}{2}P_0V_0$. Then find the value of K:

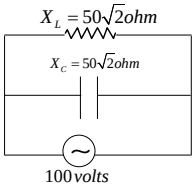


7. A source of sound of frequency 1000 Hz is moving towards a stationary wall with velocity 30 m/s and observer is also moving behind the source with the same velocity. If the frequency band width detected by the observer is $K \times 10^2$. There find the value of k:

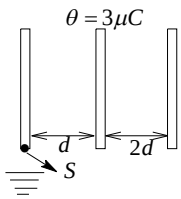
8. A vessel containing liquid of density ρ is accelerated horizontally with acceleration $10\,m/s^2$ in a gravity free space. A small sphere of density 2ρ is released from rest with respect to the vessel. Find the initial acceleration of the sphere with respect to the vessel:



9. In an AC circuit capacitive reactance of capacitor is $X_C = 50\sqrt{2}\Omega$ and inductive reactance of inductor is $X_L = 50\sqrt{2}ohm$. If the source voltage is 100 volts. Then find the current passing through the source.



10. Out of the three conducting identical plates, middle flat is given charge $3\mu C$ and other two plates are not having any charge. If switch S is closed. Find the charge passing through switch S. (in μC):



Space for Rough work

Chemistry

PART – II

SECTION – A

(One or More than One Options Correct Type)

This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

1. Dacron is a copolymer of ethylene glycol and
   (A) Phthalic acid (B) Adipic acid
   (C) Benzoic acid (D) Terephthalic acid

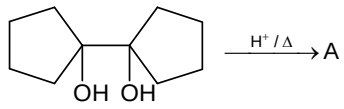
2. Select the correct statement/s to the following decay chain:
   $A \xrightarrow{-\alpha} B \xrightarrow{-\beta} C \xrightarrow{-\beta} D$
   (A) A and D are isotopes (B) B, C, D are isobars
   (C) A and B are isodiaphers (D) A and C are isotones

3. In isothermal expansion of a gas:
   (A) Kinetic energy of gas molecules remain same
   (B) Kinetic energy of gaseous molecule decreases
   (C) Pressure of gas decrease
   (D) Kinetic energy of gaseous molecule increases

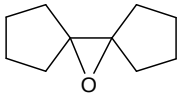
4. 5.3% (w/v) Na_2CO_3 solution and 6.3%(w/v) $H_2C_2O_4.2H_2O$ solution have same:
   (A) Molality (B) Molarity
   (C) Normality (D) Mole fraction

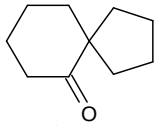
5. Given that
   $E^o_{Ni^{2+}/Ni} = -0.25V$, $E^o_{Cu^{2+}/Cu} = +0.34V$
   $E^o_{Ag^+/Ag} = +0.80V$, $E^o_{Zn^{2+}/Zn} = -0.76\,V$
   Which of the following redox process will not take place in specified direction?
   (A) $Ni^{2+}(aq) + Cu(s) \longrightarrow Ni(s) + Cu^{2+}(aq)$
   (B) $Cu(s) + 2Ag^+(aq) \longrightarrow Cu^{2+}(aq) + 2Ag(s)$
   (C) $Cu(s) + 2H^+(aq) \longrightarrow Cu^{2+}(aq) + H_2(g)$
   (D) $Zn(s) + 2H^+(aq) \longrightarrow Zn^{2+}(aq) + H_2(g)$

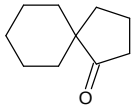
Space for Rough work

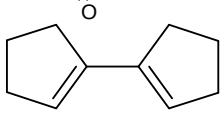
6. 

The product 'A' is

(A) 

(B) 

(C) 

(D) 

7. $CH_3CHO \xrightarrow[(ii)\ H_2O]{(i)\ CH_3MgBr} (A) \xrightarrow{H_2SO_4/\Delta} (B) \xrightarrow{HBO} (C)$

(A) and (C) are related as:

(A) Identical
(B) Position isomers
(C) Functional isomer
(D) Optical isomers

8. Select molecules in which all π-orbitals are in same plane

(A) C_6H_6
(B) $O=C=C=C=O$
(C) $CH_2=C=CH_2$
(D) $CH_2=CH-CH=CH_2$

9. Select correct statement (s)

(A) Bond length of $NO^+ > NO$
(B) Bond order of $NO^+ > NO$
(C) Bond energy of $NO > NO^+$
(D) NO is paramagnetic but NO^+ is diamagnetic

10. Which of the following properties is/are same in:

(I) $[CrCl_6]^{3-}$ (II) $[Cr(H_2O)_6]^{3+}$ (III) $[Cr(NH_3)_6]^{3+}$ (IV) $[Cr(NH_3)_6]^{3-}$

(A) Magnetic moment
(B) EAN of central metal
(C) Magnitude of splitting energy
(D) Colour

Space for Rough work

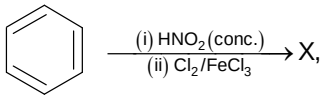
SECTION –C
Integer Answer Type

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

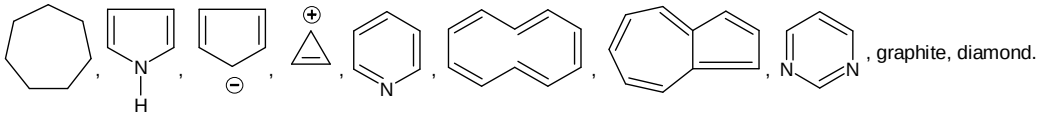
1. For the complex compound $\left[Pt(NH_3)(NO_2)Py(ONO)\right]$, total number of geometrical isomers will be:

2. Find the total number of acidic radicals which produce volatile product with dilute HCl:
$SO_4^{2-}, I^-, NO_2^-, NO_3^-, SO_3^{2-}, HCO_3^-$

3. Total number of stereoisomers of 1, 2-Dibromo-3-chlorocyclo propane.

4. 

The number of possible isomers of 'X' is?

5. Number of aromatic compounds among the following:


6. Molality of a 5M solution of H_2SO_4 having density 1.115 gm/ml.

7. Total number of electron in phosphorus which have zero value of at least one quantum number.

8. The pressure necessary to obtain 50% dissociation of PCl_5 at 400 K is numerically equal to …….times of Kp.

9. Number of S—S bonds in cyclic trimer of SO_3

10. Number of unpaired electrons in compound $[Mn(CN)_6]^{3-}$ is:

Space for Rough work

Mathematics

PART – III

SECTION – A

(One or More than One Options Correct Type)

This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

1. If x_1, x_2, x_3, x_4 are four positive real numbers such that $x_1 + \frac{1}{x_2} = 4, x_2 + \frac{1}{x_3} = 1, x_3 + \frac{1}{x_4} = 4$ and $x_4 + \frac{1}{x_1} = 1$, then
   (A) $x_1 = x_3$ (B) $x_2 = x_4$
   (C) $x_1x_2 = 1$ (D) $x_3x_4 = 1$

2. If the normals at (x_i, y_i) i = 1, 2, 3, 4 to the rectangular hyperbola xy = 2 meet at the point (3, 4), then
   (A) $x_1 + x_2 + x_3 + x_4 = 3$ (B) $y_1 + y_2 + y_3 + y_4 = 4$
   (C) $x_1x_2x_3x_4 = -4$ (D) $y_1y_2y_3y_4 = 4$

3. In a triangle ABC with sides a, b and c, a semicircle touching the sides AC and CB is inscribed whose diameter lies on AB, then the radius of the semicircle is
   (A) $\frac{a}{2}$ (B) $\frac{2abc}{s(a+b)}\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$
   (C) $\frac{2\Delta}{a+b}$ (D) $\frac{2abc}{s(a+b)}\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

4. An urn contains fair tickets with numbers 112, 121, 211, 222 and one ticket is drawn. Let A_i (i = 1, 2, 3) be the event that the ith digit of the number of ticket drawn is 1 then
   (A) $P(A_1) = P(A_2) = P(A_3) = \frac{1}{2}$ (B) $P(A_1 \cap A_2) = P(A_1)\,P(A_2)$
   (C) A_1, A_2, A_3 are pair wise independent events (D) $P(A_1 \cap A_2 \cap A_3) \neq P(A_1)\,p(A_2)\,P(A_3)$

Space for rough work

5. Suppose that the complex number z lies on the curve such that $\frac{z-4}{z-2i}$ is purely imaginary. If the complex number z_1 represents the mid-point of chord OA of this curve, O being the origin, then z_1 necessarily satisfies

(A) $\frac{z_1-2}{z_1-i} = ik$, $k \in R - \{0\}$ (B) $\frac{z_1}{z_1-2-i} = ik$, $k \in R - \{0\}$

(C) $\frac{z_1-2}{2z_1-i} = k$, $k \in R - \{0\}$ (D) $|z_1| = \frac{\sqrt{5}}{2}$

6. $\int_{-\alpha}^{\alpha} e^x + \cos x \ln\left(x+\sqrt{1+x^2}\right)dx > \frac{3}{2}$, then α can be

(A) 1 (B) 2
(C) 3 (D) 4

7. Let $f(x) = x^3 - x^2 + x + 1$ and $g(x) = \begin{cases} \max. f(t) & ; \ 0 \le t \le x \ ; \ 0 \le x \le 1 \\ 3-x & ; \ 1 < x \le 2 \end{cases}$ then

(A) g(x) is discontinuous at x = 1 (B) g(x) is continuous at x = 1
(C) g(x) is differentiable at x = 1 (D) g(x) is non-differentiable at x = 1

8. Which of the following set of values of x satisfies the equation
$2^{\left(2\sin^2 x - 3\sin x + 1\right)} + 2^{\left(2 - 2\sin^2 x + 3\sin x\right)} = 9$

(A) $x = n\pi \pm \frac{\pi}{6}$, $n \in I$ (B) $x = n\pi \pm \frac{\pi}{3}$, $n \in I$

(C) $x = n\pi$, $n \in I$ (D) $x = 2n\pi + \frac{\pi}{2}$, $n \in I$

9. Consider all the strings of n-digits which can be formed using the numbers from the set {0, 1, 2, 3} (may be starting with zero also). If E_n denotes the number of strings containing even number of zeros and O_n denotes the strings containing odd number of zeros then

(A) $E_n + O_n = 4^n$ (B) $E_n = \frac{4^n + 2^n}{2}$

(C) $O_n = \frac{4^n - 2^n}{2}$ (D) $E_n = O_n$

Space for rough work

10. The solution of $\left(\frac{dy}{dx}\right)\left(x^2y^3+xy\right)=1$ is

(A) $\frac{1}{x}=2-y^2+c\cdot e^{-y^2/2}$ (B) $\frac{1}{x}=2+y^2+c\cdot e^{-y^2/2}$

(C) $\frac{2}{x}=1-y^2+e^{-y/2}$ (D) $\frac{1-2x}{x}=-y^2+c\cdot e^{-y^2/2}$

SECTION – C

(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive)

1. Three circles lie on a plane so that each of them externally touches the other two. Two of them has radius 3, the third having radius unity. If A, B and C are the points of tangency of those circles and the area of the triangle ABC is $\frac{a\sqrt{b}}{c}$ then value of c – a – b is, where a, b and c are positive integers _____

2. The number of polynomials of the form $x^3 + ax^2 + bx + c$ which are divisible by $x^2 + 1$ where a, b, c $\in$ {1, 2, 3, ….., 10} is 10k, then k is _____

3. OABC is a tetrahedron in which O is the origin and position vector of points A, B, C are $\hat{i}+2\hat{j}+3\hat{k}$, $2\hat{i}+\alpha\hat{j}+\hat{k}$ and $\hat{i}+3\hat{j}+2\hat{k}$ respectively. An integral value of α for which shortest distance between OA and BC is $\sqrt{\frac{3}{2}}$ is _____

4. If $f(x)+f\left(\frac{1}{1-x}\right)=\frac{2(1-2x)}{x(1-x)}$, where x is a real number $x\neq 0$, $x\neq 1$, then the value of f(2) must be _____

Space for rough work

5. If $x^5 - x^3 + x = a$, when $x > 0$, then the maximum value of $2a - x^6$ is equal to _____

6. If $x^4 + 3\cos(ax^2 + bx + c) = 2(x^2 - 2)$ has two solutions with a, b, c $\in$ (2, 5) then the maximum value of $\frac{ac}{b^2}$ is _____

7. If $f(x) = (x-1)^{100}(x-2)^{2(99)}(x-3)^{3(98)} \ldots\ldots (x-n)^{100}$, let $k = \frac{f'(101)}{f(101)}$, then $\left(\frac{k}{50} - 97\right)$ is _____

8. The minimum value of $\frac{(a^2 + b^2 + c^2)}{d^2}$ in any quadrilateral, where a, b, c and d are sides of quadrilateral, is k then 6k is equal to _____

9. The number of integral values of parameter α, so that the point $(\alpha, (1 + \alpha^2)^{-1})$ does not lie outside the triangle formed by the lines $L_1 : 15y = x + 1$, $L_2 : 78y = 118 - 23x$ and $L_3 : y + 2 = 0$, is equal to _____

10. Let $f(x) = \sin x - a \sin 2x - \frac{1}{3}\sin 3x + 2ax$, increases throughout the number line then minimum prime value of a is equal to _____

Space for rough work